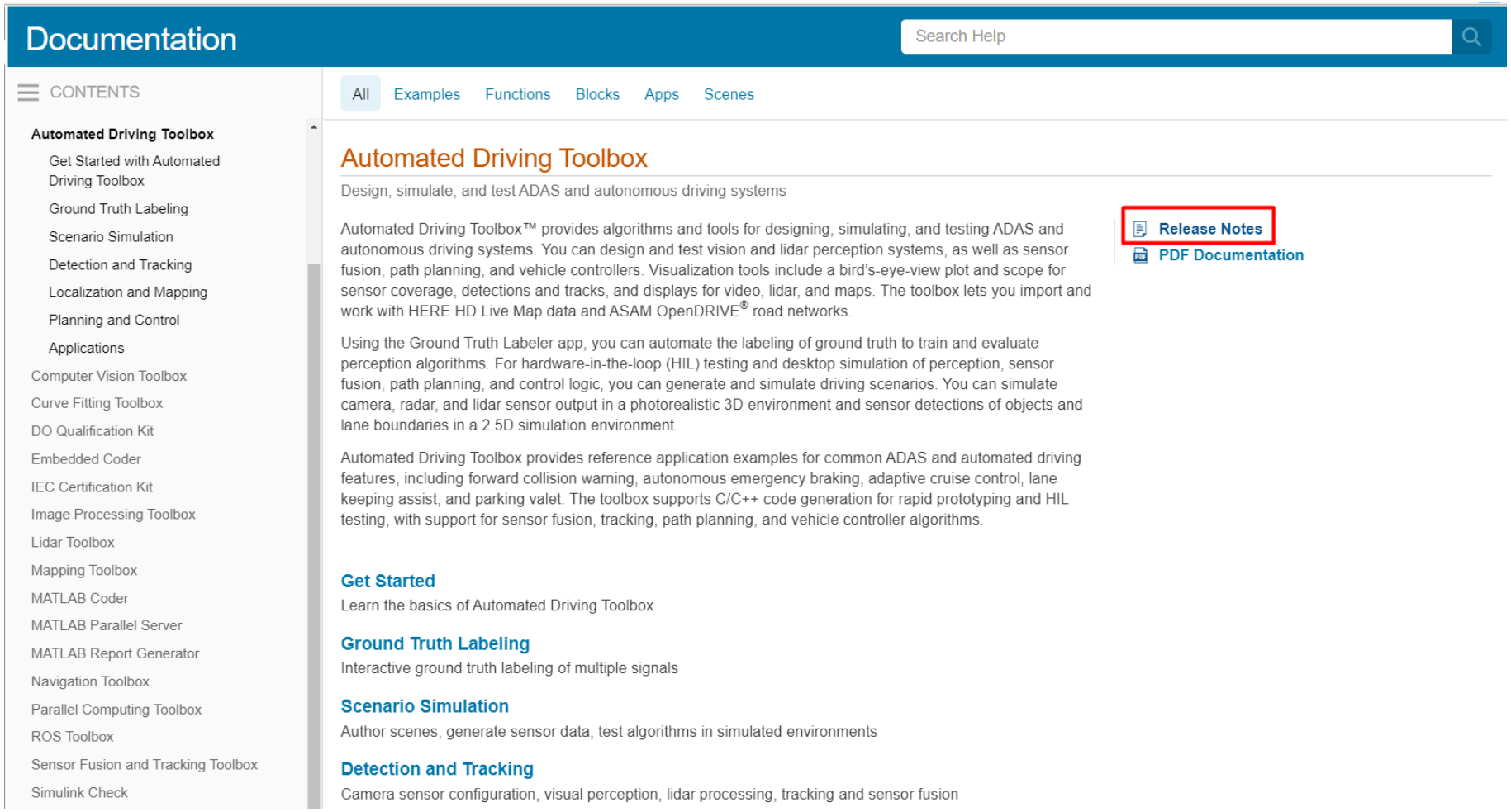
从记录的路测数据中重建驾驶场景

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21b中新加入的demo



Automated Driving Toolbox Release Notes

Bug Reports | Bug Fixes

expand all in page

Found 143 notes | Release Range: R2018b to R2021b

Sort by: Release: Latest to Earliest

R2021b

New Features, Bug Fixes, Compatibility Considerations

expand all

Ground Truth Labeling

- Labeler Enhancements: Edit cuboid ROI labels more easily in top, side, and front 2-D view projections, segment ground from lidar data using SMRF algorithm
- Velodyne Lidar Sources: Load data from Velodyne VLS-128 lidar device into Ground Truth Labeler app

Scenario Simulation

- Parking Lots: Add parking lots to driving scenarios programmatically
- ASAM OpenDRIVE Import Enhancements: Import a road network using OpenDRIVE file version V1.5 and ASAM OpenDRIVE V1.6
- ASAM OpenDRIVE Export Enhancements: Export a road network to OpenDRIVE file version V1.5 and ASAM OpenDRIVE V1.6
- ASAM OpenSCENARIO Export Enhancements: Export the routes of actors using instances of `Trajectory` element
- Scenario Reader Block: Obtain position, velocity, orientation, and acceleration information from Ego Vehicle State port
- INS Block: Generate synthetic readings from an inertial navigation and GPS sensor in driving scenarios in Simulink
- Road Heading Angles: Create more precise roads using fewer road centers
- Unreal Engine Environment Upgrade: Run 3D simulations using Unreal Engine, Version 4.25
- Position Adjustments of Unreal Engine Cameras: Update relative translation and rotation of camera sensors during simulation
- Unreal Engine Environment Performance Improvements: Run 3D simulations faster than real-time
- Unreal Engine Visualization Example: Visualize logged data for post-simulation analysis
- Lane Generation Example: Add lane information to map imported road network
- Scenario Generation Examples: Generate scenario from recorded sensor data and scenario variants from seed scenario

  The Generate Scenario from Recorded GPS and Lidar Data example shows how to automatically generate a driving scenario from the data recorded by global positioning systems (GPS) and lidar sensors. You can use the generated scenario as input data to model and simulate an automated driving system.

  The Automatic Scenario Variant Generation for Testing AEB Systems example shows how to automatically generate variants of a seed scenario in which two actors collide. You can generate random variants of a collision scenario and use them to design and validate an automated emergency braking (AEB) system.

Generate Scenario from Recorded GPS and Lidar Data

This example shows how to generate a driving scenario by extracting recorded data from global positioning systems (GPS) and lidar sensors mounted on an ego vehicle. In this example, you use prerecorded sensor data and follow these steps to generate the scenario:

Read data from the GPS sensor. Use the GPS data to:

- Import the map data from OpenStreetMap®.
- Extract the road information and driving route from the imported map data.
- Compute the waypoints for the ego vehicle.
- Estimate the ego speed and trajectory.

Read data from the lidar sensor. Use the lidar data to:

- Find the dimension and the type of non-ego actors in the driving scene.
- Estimate the entry times, exit times, positions, velocities, speeds, yaws, and trajectories of non-ego actors by using the lidar data.

Generate the driving scenario by using the extracted road network and the estimated ego and non-ego data.

This example uses:
Automated Driving Toolbox
Lidar Toolbox
Sensor Fusion and Tracking Toolbox

Open Live Script

Read GPS Sensor Data

Create a folder in the temporary directory on your computer to store the sensor data.

```
sensorData = fullfile(tempdir,"AutomotiveDataset");
if ~isfolder(sensorData)
    mkdir(sensorData)
end
```

Download the `.mat` file containing the recorded GPS sensor data and save it to a specified folder in the temporary directory. Load the GPS sensor data from the `.mat` file into the MATLAB® workspace.

```
if ~exist("data","var")
    url = "https://ssd.mathworks.com/supportfiles/driving/data/highwayLaneChangeDatase/ins2.mat";
    filePath = fullfile(sensorData,"ins.mat");
    if ~isfile(filePath)
        websave(filePath,url);
    end
    data = load(filePath);
end
```

从OSM下载GPS数据段的路网数据，将指定路段导入scenario

Driving scenario导入自车GPS，小步长仿真scenario，获取“高分辨率”自车轨迹

通过插值计算，从高分辨率自车数据中提取与激光雷达对齐的数据点

从激光雷达数据中提取track list（其他交通参与者），并将其从自车坐标系转换到scenario坐标系

导入对齐后的自车数据与其他交通参与者数据到scenario

将scenario与视频对齐播放，定性检验场景还原度